

Teen Numbers — Fixing Swapped Tens and Ones

Answer Key

Kindergarten–Grade 1

Quick Answers

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|-------|-------|-------|
| 1. 15 | 3. 17 | 5. 16 |
| 2. 13 | 4. — | 6. — |
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Common wrong answers

- If they got 51: counted the loose dots as tens and the full ten-frame as ones, so the digits came out swapped*
 - If they got 31: wrote the ones digit first, so the ones ended up in the tens place*
 - If they got 71: wrote seven-teen the way it is said, seven first, instead of tens first*
 - If they got 61: read one full ten-frame as the ones and the loose dots as the tens*
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Grading note. A swapped answer is not a counting mistake — the child counted correctly and then wrote the digits in the wrong order. The “common wrong answers” list above names the swap behind each wrong number, so the conversation can be about place value rather than about counting.

1. One full ten-frame and 5 loose dots. Write the number.

Step 1: the full ten-frame is worth ten, so there is 1 ten. Step 2: the loose dots are the ones, so there are 5 ones. Step 3: the tens digit is written first and the ones digit second, so the number is $10 + 5$, which is 15.

Watch for: a child who writes 51 has read the loose dots as tens.

2. Mia builds 1 ten and 3 ones and writes 31. Write the number her blocks really show.

Step 1: name the parts — 1 ten-stick is 1 ten, and 3 loose cubes are 3 ones. Step 2: the tens digit goes in the first (left) place and the ones digit second, so $10 + 3$ is written with a 1 first and a 3 second. Step 3: Mia wrote her 3 first, which says 3 tens and 1 one — that is why the circled digit is her 3. Her blocks really show 13.

3. Ben writes 71 for seventeen. Say why 71 is too big and write seventeen correctly.

Step 1: say the number slowly — *seventeen* means 1 ten and 7 ones. Step 2: 71 would need 7 whole ten-sticks. Seventeen has only 1 ten, so 71 is far too big; Ben wrote the number the way it is said instead of the way it is written. Step 3: put the 1 ten first and the 7 ones second: $10 + 7$, so seventeen is 17.

4. Sam builds 1 ten and 2 ones; Ty builds 2 tens and 1 one. Write $<$ or $>$.

Step 1: build Sam's number — $10 + 2$ gives twelve. Step 2: build Ty's number — $10 + 10 + 1$ gives twenty-one. Step 3: Sam has 1 ten and Ty has 2 tens, and more tens always wins, so Sam's number is the smaller one: $<$.

Teaching point: the same two digits, swapped, are not the same number.

5. Ana counts one full ten-frame and 6 more dots and writes 61. Find her mistake and write the true number.

Step 1: count the frames — the first frame is full (1 ten) and the second frame holds 6 dots (6 ones). Step 2: Ana's mistake is that she wrote the 6 first. A 6 in the first place means 6 whole ten-frames, and there is only one full frame on the page. Step 3: 1 ten and 6 ones is $10 + 6$, so the dots show 16 .

6. Jo swaps the digits of the number made from 1 ten and 8 ones. Compare Jo's swapped number with the number she meant.

Step 1: the number Jo meant is $10 + 8$, which is eighteen — 1 ten and 8 ones. Step 2: swapping the digits puts the 8 in the tens place, giving 8 tens and 1 one, which is eighty-one. Step 3: 8 tens is far more than 1 ten, so Jo's swapped number is the bigger one: $>$. The digit sitting in the tens place of Jo's swapped number is the 8.